

PlusOrders Company Data Mart

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Mestrado em Engenharia

Informática

Armazém de Dados

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Data: 16/11/2024

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1. Introduction

This report presents the **Part 1** deliverable of the Data Warehouse project for the PlusOrders company, as described in the assignment. The objective is to design the architecture, dimensional model, staging structures, data mapping, SCD strategy, and cleaning/transformation logic for a data mart supporting analytics on customer orders.

PlusOrders operates internationally, collecting all order, customer, product, and site information into a central transactional database. The company intends to build a data mart optimized for historical analysis of orders, customer behavior, sales performance, and product availability across multiple countries and currencies.

To achieve this, we applied the **Kimball methodology**, adopting a **bottom-up, dimensional modeling** approach and focusing on the granularity that provides maximum analytical flexibility: **the individual order-line item**.

2. Data Warehouse Architecture

The proposed architecture consists of four layers, each with specific responsibilities:

2.1 Source Systems Layer

The source is the Plus Orders operational database containing:

- Orders, order lines, order line details
- Products and product videos
- Customers
- Countries
- Currencies
- Sites, site info
- Boxes (shipping packaging)
- Sizes
- Timestamps for creation and updates

These tables provide richly detailed data, but contain inconsistencies, duplicates, non-standardized values and foreign-key integrity issues.

2.2 Staging Area (SA)

The **Staging Area** is designed to accommodate:

A) Raw tables (1:1 copies of source)

Used for profiling and comparisons for incremental loads.

B) Cleaned tables

Contain validated, transformed and deduplicated records.

C) Reject tables

Records failing quality rules (unfixable) are not loaded into the data mart.

D) Slowly Changing Dimension detection

Staging retains previous loads to identify attribute changes.

E) Incremental load support

Only new or updated rows are processed.

The entire staging environment is free of business logic and intended for temporary ETL operations.

2.3 ETL / Integration Services Layer

Implemented using **SQL Server Integration Services (SSIS)**, based on the following design principles:

- **Efficient extraction** with no full reloads — only deltas.
- **Data profiling** before transformation to detect anomalies.
- **Cleaning and standardization** (e.g., date formats, text normalization, numeric type validation).
- **Duplicate detection** using exact and approximate matching (products especially).
- **Validation of referential integrity** and rejection of invalid rows.
- **SCD Type 1/2 processing** for dimension updates.

- **Currency conversion** using daily exchange rates.

All paths and connection managers are configured via an **external configuration file** as required.

2.4 Data Mart Layer (DW)

The final Data Mart uses a **Star Schema** to support fast OLAP queries.

Key characteristics:

- **Grain:** *one row per order-line item*
- **Normalized currency:** values in both *local currency* and *Euro*
- **Daily currency exchange fact table**
- **Dimensional hierarchies:** date, product → category, site → country, customer → geography
- **Support for drill-down & roll-up** across all business hierarchies

This structure satisfies all business questions, including those involving time frames, categories, geographic regions, shipping costs, and currency harmonization.

3. Dimensional Modeling (Star Schema)

3.1 Fact_OrderLines (Main Fact)

Grain:

One row per product per size per per costumer per site per country per category per box per currency

Foreign Keys

- date_key
- customer_key
- product_key
- size_key
- site_key
- country_key
- category_key
- box_key
- currency_key

Measures

- quantity
- line_total_local_currency
- line_total_eur
- allocated_shipping_cost (see below)
- discount_applied
- promotion_discount

Shipping Cost Allocation

Since shipping cost is stored at order-level:

$\text{AllocatedShipping} = \text{Order.ShippingCost} / \text{SUM}(\text{quantity in same order})$

Attributes

Attribute

Type

order_line_key	IGINT
order_id_source	BIGINT
date_key	INT
customer_key	BIGINT
product_key	BIGINT
size_key	BIGINT
site_key	BIGINT
country_key	BIGINT
category_key	BIGINT
box_key	BIGINT
currency_key	BIGINT
quantity	INT
line_total_local_currency	DECIMAL(18,4)
line_total_eur	DECIMAL(18,4)
shipping_cost_allocated	DECIMAL(18,4)
discount_amount	DECIMAL(18,4)
promotion_discount	DECIMAL(18,4)

3.2 Fact_CurrencyExchange

Grain:

One row per currency per day

Purpose:

Provides the daily EUR exchange rate needed to normalize multi-currency analysis.

Measures:

- exchange_rate_to_eur

Used to compute:

$\text{line_total_eur} = \text{line_total_local_currency} * \text{exchange_rate_to_eur}$
 $\text{shipping_cost_eur} = \text{allocated_shipping_cost} * \text{exchange_rate_to_eur}$

Attributes:

Attribute	Type
SourceCurrencyKey	BIGINT
DestinationCurrencyKey	BIGINT
DateKey	BIGINT
SourceDestinationExchange	DECIMAL(18,8)
DestinationSourceExchange	DECIMAL(18,8)

3.3 Dimensional Tables + Attributes

Below are the dimensions included in the model and justification for their SCD type.

3.3.1 Dim_Date (Type 0 / Static)

Pre-populated date dimension with several data to support business for, monthly totals, daily totals, quarterly roll-ups, weekend-specific analysis, etc

Attribute	Type
date_key	INT
full_date	DATE
day	SMALLINT
month	SMALLINT
month_name	VARCHAR (20)
quarter	SMALLINT
year	SMALLINT
week	SMALLINT
is_weekend	BOOLEAN
day_of_year	SMALLINT
fiscal_year	SMALLINT

3.3.2 Dim_Product (SCD Type 2)

Justification for Type 2:

The descriptions of the products might change or update over time, products can appear duplicated in the source, and it requires historical tracking

Attribute	Type
product_key	BIGINT
product_id_source	BIGINT
description	TEXT
category_key	BIGINT
available_portal	SMALLINT
box_key	BIGINT
active	BOOLEAN
valid_from	TIMESTAMP
valid_to	TIMESTAMP
is_current	BOOLEAN

3.3.3 Dim_Category (SCD Type 1)

Category changes are rare and should overwrite previous values.

Attribute	Type
category_key	BIGINT
category_id_source	BIGINT
name	VARCHAR(255)
gender	VARCHAR(20)

3.3.4 Dim_Customer (SCD Type 2)

Customer attributes such as geographic region, language, or VIP status may change and must be tracked historically.

Attribute	Type
customer_key	BIGINT
customer_id_source	BIGINT
country_key	BIGINT
date_of_birth	DATE
gender	VARCHAR(10)
geographic_region	INT
language	VARCHAR(50)
vip_customer	BOOLEAN
valid_from	TIMESTAMP
valid_to	TIMESTAMP
is_current	BOOLEAN

3.3.5 Dim_Country (SCD Type 1)

Attributes like VAT or Euro zone membership rarely change and should overwrite.

Attribute	Type
country_key	BIGINT
country_id_source	BIGINT
country_name	VARCHAR(255)
country_code	VARCHAR(10)
vat	DECIMAL(6,4)
euro_zone	BOOLEAN

3.3.6 Dim_Site (SCD Type 1)

SCD1 because changes overwrite.

Attribute	Type
site_key	BIGINT
site_id_source	BIGINT
site_name	VARCHAR(255)
site_url	VARCHAR(512)
site_initials	VARCHAR(20)
site_currency	VARCHAR(3)

3.3.7 Dim_Box (SCD Type 1)

Shipping boxes rarely change, and historical tracking is unnecessary.

Attribute	Type
box_key	BIGINT
box_id_source	BIGINT
box_name	VARCHAR(255)
box_height	DECIMAL(10,4)
box_length	DECIMAL(10,4)
box_width	DECIMAL(10,4)
box_vol_weight	DECIMAL(12,4)

3.3.8 Dim_Size (SCD Type 1)

The original size table contains many irrelevant columns (_17, _18, _19... etc.), which are discarded.

Attribute	Type
size_key	BIGINT
size_id_source	BIGINT
size_name	VARCHAR(50)
friendly_name	VARCHAR(100)
min_size	SMALLINT
max_size	SMALLINT

3.3.9 Dim_Currency (SCD Type 1)

Attribute	Type
currency_key	BIGINT
currency_code	VARCHAR(3)
currency_name	VARCHAR(100)
currency_symbol	VARCHAR(5)

4. Staging Area Structures

The staging area includes three categories of tables:

4.1 Raw Tables

Exact copies of source tables:

- SA_raw_orders
- SA_raw_orders_lines
- SA_raw_orders_lines_details
- SA_raw_products
- SA_raw_customers
- SA_raw_sizes
- SA_raw_countries
- SA_raw_currencies
- SA_raw_boxes
- SA_raw_sites
- SA_raw_sites_info

Purpose:

- retain original data
- support incremental delta detection
- support profiling and audit

4.2 Cleaned Tables

These tables contain data after:

- standardization
- reference validation
- type normalization
- deduplication
- removal of unverifiable rows

4.3 Reject Tables

Records with irrecoverable issues are stored here.

Examples of rejection criteria:

- missing mandatory fields (product_id, order_date, currency_id, etc.)
- invalid foreign keys
- negative totals
- invalid date formats
- corrupted numeric fields
- duplicate rows with conflicting values

These are excluded from the warehouse.

5. Data Mapping

This section maps source → staging → warehouse.

Also please review the file Mapping for a more visual representation of the mapping

5.1 Orders

Source	Staging	DW
order_date	validated timestamp	date_key
shipping_cost	numeric check, >= 0	allocated across lines
currency_id	FK validated	currency_key
country_id	validated	country_key
customer_id	validated	customer_key

5.2 Order Lines

Source	Staging	DW
product_id	deduped/fuzzy match	product_key
quantity	numeric validation	quantity
line_total	numeric validation	line_total_local_currency
size_id	validated	size_key

5.3 Order Line Details

Discount values are merged into the fact.

5.4 Products

Source	Transformation	DW
description	trimmed, cleaned	description

category_id	FK validated	category_key
active	normalized to 0/1	active
duplicates	merged using fuzzy logic	product_key

5.5 Customers

Attributes are mapped directly and SCD2 logic inserted into Dim_Customer.

5.6 Sizes

Only key fields retained.

5.7 Currency Exchange

External or company-provided rate tables loaded into Staging.

Mapped to:

- date_key
- currency_key
- exchange_rate_to_eur

6. Slowly Changing Dimensions (SCD)

Type 1 Dimensions:

- Dim_Category
- Dim_Country
- Dim_Site
- Dim_Box
- Dim_Size
- Dim_Currency

Justification:

Changes should overwrite historical values.

Type 2 Dimensions:

- Dim_Product
- Dim_Customer

Justification:

- Product descriptions, assignments, and activity status change historically.
- Customer demographics (region, language, VIP status, gender) also change over time.
- Business questions require historically accurate attribute context.

Each SCD Type 2 table includes:

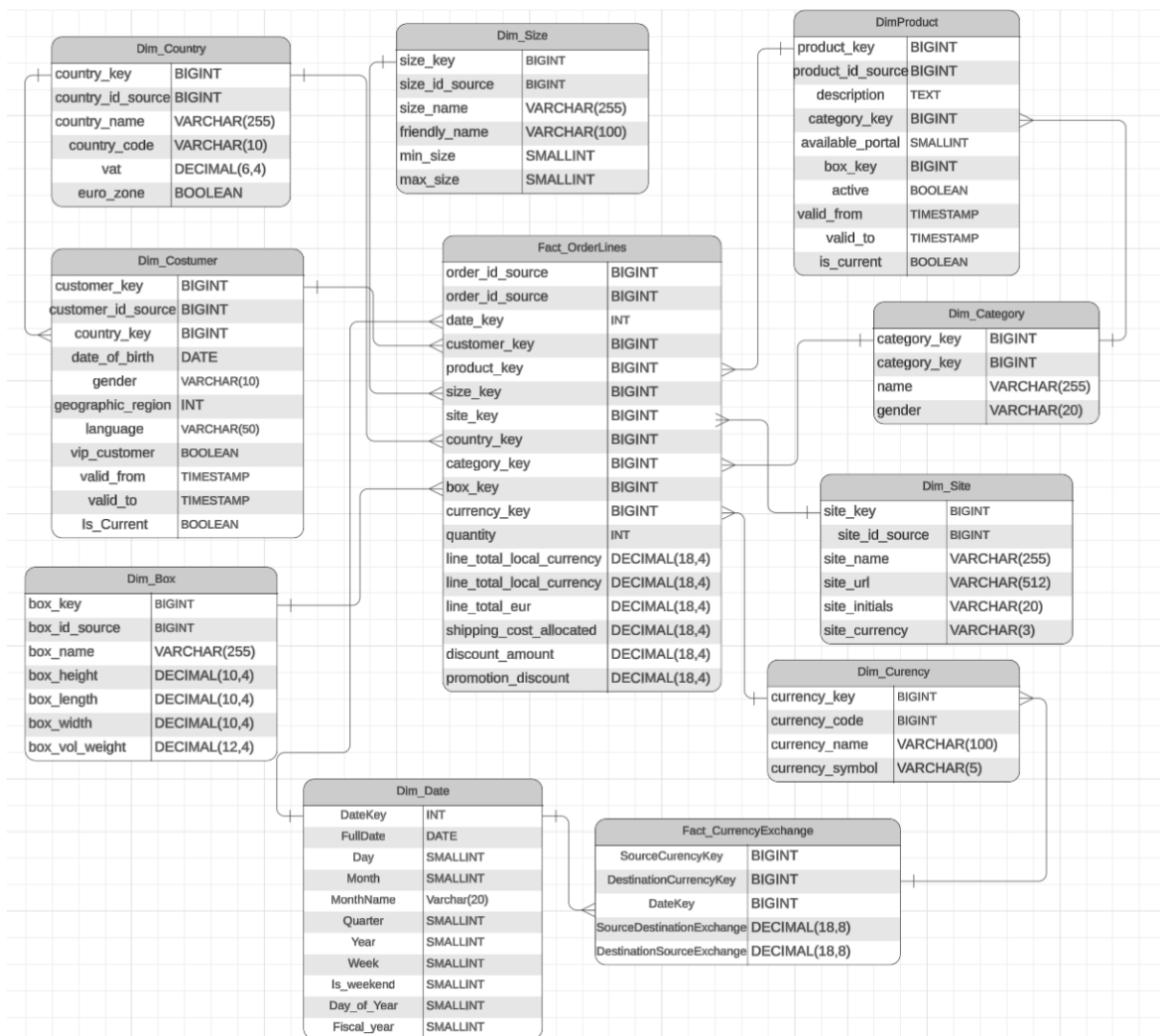
- valid_from
- valid_to
- is_current

7. Final Star Schema

Fact_OrderLines at the center surrounded by:

- Dim_Date
- Dim_Product
- Dim_Customer
- Dim_Site
- Dim_Country
- Dim_Size
- Dim_Currency
- Dim_Box
- Dim_Category

Also, the Fact_CurrencyExchange is linked to the star schema



8. Conclusion

The dimensional model, staging area, SCD strategy, and ETL design presented in this report fully satisfy the functional and technical requirements of PlusOrders.

The use of a **granular Fact_OrderLines** enables high analytical flexibility, while the introduction of **Fact_CurrencyExchange** allows reliable cross-currency analysis in EUR.

This architecture ensures:

- high performance
- historical accuracy
- clean and reliable data
- support for multi-level drilldowns and roll-ups
- extensibility for future data sources and business needs

This completes **Part 1** of the project.